

PAVAN RAJ R

Software Developer | AI/ML Engineer | Computer Vision

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PROFESSIONAL SUMMARY

Software Developer and AI/ML Engineer with hands-on experience building a full-stack CRM (React, FastAPI, MySQL, Docker) and an industrial computer vision system at Robert Bosch. Skilled in Python, OpenCV, TensorFlow and PyTorch. Focused on building reliable, production-ready software that reduces manual work and errors.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript

Web & Backend: React, FastAPI, REST APIs, MySQL, Docker, Git, GitHub

AI/ML: TensorFlow, PyTorch, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn, NLP, Google Gemini API

Computer Vision: OpenCV, DeepFace, LBPH, Template Matching, Pypylon SDK

Data & BI: Power BI, DAX, SQL, MySQL

Tools: Jupyter, PyCharm, VS Code, Selenium WebDriver, Tkinter, PyQt5

WORK EXPERIENCE

Software Developer | Sri Swastik Services

Sept 2026 - Present

- Developing a role-based Debt Recovery CRM (in development) with four access levels: Super Admin, Admin, Team Lead and Agent.
- Building features for case allocation, predictive and manual dialing, call logging, attendance and leave tracking, and analytics dashboards.
- Designed and deployed the company's new website.
- Tech stack: React, FastAPI, MySQL, Docker.

AI/ML & Computer Vision Intern | Robert Bosch BidP, Bangalore

Jan 2026 - May 2026

- Built a vision system that automatically inspects glow plugs, reducing manual inspection errors by ~85%.
- Processed live camera images with OpenCV and cut false detections by 30%.
- Created a simple desktop app (Tkinter) used by operators on the factory floor for live inspection.
- Saved every PASS/FAIL result in timestamped CSV files for easy tracking.

PROJECTS

Glow Plug Inspection System | Python, OpenCV, NumPy, Tkinter

- Automated inspection from image capture to PASS/FAIL logging; improved detection accuracy by 40% over standard methods.

AI Voice Assistant | Python, Google Gemini API, SpeechRecognition, PyQt5

- Voice assistant that listens continuously, gives context-aware answers using the Gemini API, and automates system tasks.

Face Recognition & Emotion Detection | Python, OpenCV, DeepFace, LBPH

- Live webcam system that recognizes faces and classifies emotions; new users can be added without retraining the model.

Power BI Sales Dashboard | Power BI, DAX

- Interactive dashboard tracking revenue, churn and KPIs by region and product; cut monthly reporting time by 60%.

EDUCATION

Master of Computer Applications (MCA) - AI, ML & Data Science

2024 - 2026

S-VYASA Deemed University, Bangalore

Bachelor of Science (BSc) - Physics & Computer Science

2022 - 2024

Surana Autonomous College, Bangalore

CERTIFICATIONS

- GenAI Powered Data Analytics - Tata Consultancy Services (Forage), May 2026
- Operations Industrial Engineer Job Simulation - Siemens (Forage), May 2026